Special Notice (SN) ARPA-H-SN-26-150

Proposers' Day Announcement for
Advanced Research Projects Agency for Health (ARPA-H)
Health Science Futures (HSF) Office
Opportunity: Systematic Targeting Of MicroPlastics (STOMP)

I. KEY INFORMATION

Table 1. Proposers' Day Key Information	
Proposers' Day	Wednesday, April 22nd, 2026, from 8:30 AM to 5:00 PM ET
Location	Washington, DC (Hybrid)
Website for registration	https://solutions.arpa-h.gov/STOMP
Registration opens	April 2nd, 2026, at 2:00 PM ET
Registration closes	In-person: Friday, April 17, 2026, at 12:00 PM ET Virtual: Tuesday, April 21, 2026, at 12:00 PM ET
Technical Points of Contact	Shannon Greene (TA2) and Ileana Hancu (TA1), Program Managers, ARPA-H / HSF Office
Program email	STOMP@arpa-h.gov
Proposer deliverables, details below	In person - Provide one (1) physical poster (to fit within a space measuring 3.75 feet wide by 3.75 feet high – landscape layout preferred STOMP program page: Submit one (1) PDF page "teaming profile" (proposer's capabilities, composition, research areas of interest, etc.)

II. INTRODUCTION

This Special Notice (SN) is intended to alert the scientific community to a newly approved ARPA-H Program and notify potential proposers of an associated Proposers' Day (PD). Under the cognizance of HSF Mission Office, ARPA-H is publishing an Innovative Solutions Opening (ISO) for the STOMP program no later than April 2nd, 2026.

The HSF Mission Office of the ARPA-H is sponsoring a PD for the planned proposer community in support of the Systematic Targeting Of MicroPlastics (STOMP) program.

Potential proposers are encouraged to attend the optional Proposers' Day. While not mandatory, ARPA-H is hosting the STOMP PD to achieve the following objectives:

- Draw attention of the scientific community to the STOMP technical areas.

- Provide information about the program to potential proposers, including details about the overall program goals, solution summary, and full proposal structure for the forthcoming ISO, and how they might respond to ARPA-H's forthcoming ISO.
- Share the intended scope, timeline, and scale of the STOMP program and how the proposers, in collaboration with the ARPA-H team and resources, can accomplish the collective mission of accelerating healthcare outcomes for all Americans.
- Initiate teaming arrangements among potential proposers to achieve program goals.

The PD will include brief overview presentations by government personnel, a brainstorming session on potential TA3 solutions (solicited later in the program), and posters for potential proposers to highlight technical capabilities and promote teaming arrangements. ARPA-H will collect participant questions throughout the event and publish answers to a publicly available Frequently Asked Questions (FAQ) in: https://arpa-h.gov/research-and-funding/programs/STOMP/faqs and on https://sam.gov/

Attendance at this event is not a requirement for submission of a solution summary, full proposal, or selection for funding; however, it is anticipated that unique multi-party teaming arrangements will be needed to successfully innovate and integrate the critical technologies necessary to meet the STOMP program goals. Information relayed during the PD will be made available on the HSF Office section of the ARPA-H Opportunities page: https://arpa-h.gov/research-and-funding/programs/STOMP as well as SAM.gov.

Advance Registration is required. Registration requests will only be accepted from the potential proposer community as participation is not intended for patients, patient advocates, media or a general interest audience.

III. STOMP– PROGRAM INFORMATION

While full details of the STOMP program will be provided in the forthcoming STOMP ISO, the following is a brief description of the full program. Proposers should refer to the forthcoming STOMP ISO for more details.

Overview

Plastics are ubiquitous in modern life, and the amount of plastic in the world increases every year. With plastic comes plastic waste, and with plastic waste, microscopic plastics, or microplastics. A growing body of evidence suggests that micro- and nano-plastics (MNPs) may harm human health. The STOMP program seeks to better quantify MNPs in humans; understand the mechanisms of harm; and, ultimately, improve human health by developing means to limit uptake and remove MNPs from the body.

Recently, the scientific community has investigated the impact of microplastics on human health, with many studies showing *correlation* between microplastics presence and deleterious health effects. While correlation does not demonstrate causation, causation is difficult to prove in humans. Exposure studies in animals, however, demonstrate that MNP exposure *causes* harm. Despite growing and concerning evidence, questions about microplastics-related harm remain. Transit at systemic and cellular scales is not clear: how do microplastics get into organ systems? How do they get into cells? Where do they accumulate within cells? Is there an innate removal mechanism? Is there an equilibrium between the blood and organ levels? Toxicity mechanisms are only now beginning to be elucidated: which biochemical pathways do microplastics dysregulate, and how? How do microplastics dysregulate the microbiome? How do microplastics disrupt tissue structure and function? Furthermore, it is still not clear which microplastics are most harmful, at what dose and why. Is it the size, shape, or chemical composition—or all the above?

There are many reasons why these questions remain unanswered. First, measuring microplastics in biological tissue is very challenging. Existing technologies require intensive sample preparation and data analysis and/or imaging workflows that can take hours to days, with limited standardization across laboratories. No single analytical technique can currently determine microplastic particle mass, count, size and chemical composition simultaneously, nor is there a standard workflow for labeling and tracking microplastics *in vivo*. Furthermore, the biological matrix— the complex mixture of proteins, salts, lipids, and other materials that comprise a biological sample— interferes significantly with measurement. In addition, experiments are often performed with animals exposed to varying, and sometimes unreasonably high, MNP levels, which do not recapitulate human exposure. Last, but not least, there are no good MNP reference materials enabling scientists to perform experiments that can be directly compared to each other. Polystyrene spheres are used in most studies, as they are one of the few commercially available solutions, even though it is generally agreed upon that this is not a good model of human exposure. Fibers and irregular fragments produce stronger biological effects than spheres, because they are retained longer and cause more direct membrane damage, while also potentially triggering stronger innate immune activation.

Technical Approach

The STOMP program will develop 1) reliable measurement tools for MNPs in biological samples, 2) a mechanistic framework for MNP toxicity and 3) solutions for removal of MNPs from the human body.

To accomplish these goals, STOMP has three (3) technical areas (TAs):

TA1: Measurement

Without reliable detection, characterization, and quantification of microplastics, it is not possible to understand where MNPs are in the human body (and why/how), nor is it possible to validate removal methods. STOMP TA1 performers will perform three (3) activities:

a. Develop a best-in-class research-grade MNP measurement technique, including sample preparation, measurement, and analysis methodology, to be used by STOMP teams to quantify data and validate results starting before the end of the first year of the STOMP period of performance. Because of the accelerated timeline, performers will likely develop techniques that overcome challenges with existing instrumentation.
b. Develop a novel, clinically focused measurement system capable of quantifying MNPs in human biospecimens quickly, inexpensively, and at scale. This may involve the development of entirely new devices and modalities for rapid MNP detection and quantification.
c. Conduct a study in human participants, acquiring paired MNP data from biological fluids and tissue samples, to determine if tissue MNP concentration correlates with the MNP concentration in the more accessible biological fluids.

TA2: Understanding

To target the most harmful microplastics in the most effective way, it is necessary to understand which microplastics drive toxicity, and how. TA2 performers will:

a. Conduct acute and chronic MNP exposure studies in animal models to assess bioaccumulation and impacts to system- and organ-level harm.
b. Prioritize organ systems with the greatest MNP accumulation and indications of harm to investigate and define trafficking and accumulation mechanisms to key organ systems.
c. Determine cell types and cellular pathways impacted by MNPs, including outlining when intracellular internalization occurs and mechanisms of toxicity (oxidative stress, inflammation, protein dyshomeostasis, etc.).
d. Identify any innate mechanisms for MNP removal and degradation.

TA3: Removal

The ultimate aim of STOMP is to develop methods to remove toxic microplastics from the human body, to improve health outcomes. Although **TA3 solutions will not be solicited until Month 24**, it is anticipated that TA3 performers will develop and preclinically validate technologies that prevent further accumulation and reduce microplastic burden.

Proposal Options

Proposers submit separate proposals for TA1 and TA2. TA1/TA2 performers must address *all* components of TA1/TA2, respectively; teaming will likely be necessary. ARPA-H reserves the right to select only certain components of TA1 or TA2 submitted by the same performer team and pair performers to complete TA1 and/or TA2 together. Performers must commit to working with the partners selected by ARPA-H, in the case that only certain sub-areas are selected from one team's proposal. **ARPA-H is not accepting TA3 proposals until later into the program. TA3 proposals submitted alongside TA1 and TA2 will not be reviewed.** A Special Notice and amended ISO for TA3 will be released upon completion of initial TA1 and TA2 Phase 1 studies.

Program Structure

STOMP is structured into two (2) 30-month phases:

Phase 1 (0-30 months) (1) Develop and validate best-in-class research-grade quantification techniques for MNPs in biological samples, develop prototype clinical assays for MNP measurement, and complete a clinical study of MNP levels across biofluids and tissue samples; and (2) Define key organ systems affected by the most toxic polymer types and identify mechanisms of internalization and toxicity.

Phase 2 (31-60 months) (1) Further develop and validate MNP clinical assay and (2) Demonstrate therapeutic efficacy of MNP prevention, degradation or clearance solutions in preclinical models, while using the results of Phase 1 work (research-grade measurement tools and mechanisms of MNP internalization and toxicity).

Solicitation and selection of TA3 performer teams will take place close to the completion of Phase 1 for TA1 and TA2, dependent on the completion of technical metrics, and the PIs and ARPA-H will convene in yearly meetings in addition to regular monthly reporting requirements.

Phase 1					Phase 2				
6	**12**	**18**	**24**	**30**	**36**	**42**	**48**	**54**	**60**
TA1: Measure									
TA2: Understand									
				Apply to TA3	**TA3: Remove**				

Figure 1. STOMP Program Structure and General Overview

IV. STOMP – PROPOSERS' DAY INFORMATION

Proposers' Day Participation and Deliverables

This SN and PD include two specific opportunities for prospective proposers to identify team members:

Posters

Proposer's Day will include an optional poster session designed to facilitate collaboration and team formation. Interested attendees and/or proposers are invited **but not required** to request space for one poster per team that summarizes the research interests and/or technical capabilities related to the therapeutic area(s) that the team plans to propose. A detailed style guide and further directions pertaining to the poster will be provided upon your registration acceptance, but posters should measure no larger than 3.75 ft x 3.75 ft, and landscape layout is preferred. At PD, posters will be grouped into TA1 or TA2 categories. Presenters will have one hour to stand by the poster, present their work, and answer questions from interested parties. Posters will be accepted on a first-come, first-served basis, until the maximum possible number of submissions given the space constraint for the PD is reached, considering relevant and diversity of presenters. Interested attendees must identify their intention to present a poster during registration. Poster presentations are restricted to those attending the meeting.

Teaming Profiles

Interested parties are also invited to submit a 'teaming profile' via the link here (https://solutions.arpa-h.gov/Teaming) and below. The teaming profile will describe the technical competencies, team capabilities, team composition, research areas of interest, unique facilities and other capabilities, as they relate to the program, and desired technical/other competencies sought from other potential team partners. The profile will include, at a minimum:

- Contact information, including name, organization, email, telephone number, mailing address, and website

- Brief description of the proposer's technical competencies and relevant facilities
- Desired technical competencies and facilities from other potential team members, if applicable

All teaming profiles must be submitted via the STOMP Teaming Profile Form Link (https://solutions.arpa-h.gov/Teaming). The teaming profile will be updated regularly and made available on the program's teaming profile site (https://arpa-h.gov/research-and-funding/programs/STOMP/teaming). Specific content, communications, networking, and team formation are the sole responsibility of participants. ARPA-H does not endorse any participating organization or exercise any responsibility for improper dissemination of the team profiles.

Live Q&A

A live Q&A session will be led by the Program Managers and Business Innovation Division (BID) during STOMP PD. STOMP PD participants are encouraged to use this time to discuss any STOMP contracting questions you have with our contracting experts. ARPA-H will collect participants' questions and BID responses and make them available in the Frequently Asked Questions (FAQ) in:
https://arpa-h.gov/research-and-funding/programs/STOMP/faqs and on Sam.gov.

TA3 Brainstorming Session

Although STOMP is not soliciting TA3 proposals until approximately Month 24 of the program, interested TA3 proposers are encouraged to attend and participate in a TA3 brainstorming session. Participants will break into small groups to discuss potential strategies for (1) prevention of MNP adsorption and trafficking in the body; (2) degradation of MNPs into non-toxic byproducts; and (3) expulsion of MNPs from cells and tissues for excretion.

Registration Information

Participants must register in advance through the registration website. The registration will remain open until 48 hours before the start of the event. There is no registration fee for the STOMP Proposers' Day. Note that there will be **no same-day registration**. An online registration form and various other meeting details can be found at the registration website:
https://solutions.arpa-h.gov/STOMP.

All inquiries should be addressed to STOMP@arpa-h.gov. Please refer to the STOMP Proposers' Day (ARPA-H-SN-26-150) in all correspondence. Prior to submitting an e-mail inquiry, please check the General Program FAQ's (https://arpa-h.gov/research-and-funding/programs/STOMP/faqs) or sam.gov to see if your question is answered there.

V. POINTS OF CONTACT

Technical Point of Contact

TA1: Ileana Hancu, Program Manager, ARPA-H / HSF Office

TA2: Shannon Greene, Program Manager, ARPA-H / HSF Office
Email: STOMP@arpa-h.gov
All inquiries should reference “STOMP Proposers’ Day (ARPA-H-SN-26-150) in the subject line of the email.

VI. WEBINAR INFORMATION

Virtual attendees will receive the link to the webinar once they are verified and registration is confirmed.

VII. DISCLAIMER

This SN is issued solely for information and potential new program planning purposes; the SN does not constitute a formal solicitation for proposals or solution summaries. Responses to this notice are not offers and cannot be accepted by the Government to form a binding award. Submission of any information in response to the SN is voluntary and is not required to propose to subsequent ARPA-H ISOs or research solicitations on this or any other topic. ARPA-H will not provide reimbursement for costs incurred in responding to this SN. Respondents are advised that ARPA-H is under no obligation to acknowledge receipt of the information received or provide feedback to respondents with respect to any information submitted under this SN.